

Handy Helper Umbrella

8-Panel Field Reference for HVAC & Electrical Trades

1 Safety First

LOTO, PPE & Wet-Work Precautions

Lockout/Tagout (LOTO)

1. Notify anyone affected
2. Shut down by normal means
3. Isolate — disconnect/break energy source
4. Lock out — apply individual lock + tag
5. Verify de-energized — test meter on known source first, then circuit
6. Release — only the person who applied the lock removes it

Wet-Work Precautions

Never pressure wash energized electrical components. Cover motors, contactors, control boards. Keep water away from disconnects/panels. Allow equipment to dry before re-energizing. Use GFCI-protected power.

PPE Checklist

Safety glasses/face shield | Electrical-rated gloves (Class 0) | Hearing protection | EH-rated boots | Long sleeves for brazing

OSHA Lockout/Tagout (29 CFR 1910.147)

2 Refrigeration Cycle

The 4 Stages of the Cycle

The 4 Stages

1. Compressor — raises pressure & temperature of vapor
2. Condenser — rejects heat, vapor becomes liquid
3. Metering Device — drops pressure, liquid becomes mixed phase
4. Evaporator — absorbs heat, refrigerant becomes vapor

COMPRESSOR

▲ ▼

Low-P High-P
vapor vapor

▲ ▼

EVAP CONDENSER

▲ ▼

←■ METERING DEVICE ■→

Pressure Zones (R410A approx)

High side: 350-450 PSIG, 100-120°F saturated

Low side: 100-140 PSIG, 40-55°F saturated

Metering Devices: TXV (best efficiency) | Fixed piston (charge-sensitive) | Cap tube (small systems) | EEV (electronic, variable)

HVAC School — Refrigeration Cycle

Anthony's Cool School HVAC Training

3 Superheat & Subcooling

Charge & Performance Diagnostics

Superheat = Suction Line Temp – Sat. Suction Temp
Measures evaporator performance

Superheat	Likely Cause
High (>15-20°F)	Undercharge, restriction, low airflow
Low (<5-10°F)	Overcharge, overfeeding, high airflow
Normal (8-14°F)	System OK (TXV: 10-14, fixed: 8-12)

Subcooling = Sat. Liquid Temp – Liquid Line Temp
Measures condenser performance

Subcooling	Likely Cause
High (>15-20°F)	Overcharge, liquid line restriction
Low (<5-10°F)	Undercharge, condenser airflow issue
Normal (8-12°F)	System OK

Steps: Measure line temps | Read gauges | PT chart for sat temps | Calculate | Compare to mfr specs

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4 Thermostat Wiring

Low-Voltage Terminal Reference

Terminal Reference

Term	Color	Function
R/RH/RC	Red	24V power (heat/cool)
C	Blue	Common return
Y/Y1	Yellow	Cooling → contactor
G	Green	Fan / blower
W/W1	White	Heating
O	Orange	Rev. valve — cool (most)
B	Dk Blue	Rev. valve — heat (Ruud/Rheem)
W2	White	Aux / 2nd-stage heat

Common Failures

No 24V at stat: Check transformer, fuse, float switch
Stat blank: Check R-C for 24V, C wire connection
Fan won't stop: G wire shorted, relay stuck
No cooling: Y not reaching contactor, float/pressure switch
Rev. valve wrong: Check O/B — energize-in-cool vs heat
Float switch: Breaking R kills stat + defrost board. Breaking Y kills cooling only.

HVAC School — Low Voltage Wiring

5 Contactor & Capacitor

Motor Circuit Testing

Contactors

Coil: 24V from stat Y → safety switches → contactor coil

Contacts: 240V through contacts to compressor + condenser fan

Test: Click sound, 24V across coil, ~0V across closed contacts

Capacitor Terminals

Terminal	Connects To
C (Common)	Run winding + power
HERM	Compressor
FAN	Condenser fan motor

WARNING: Capacitors hold charge even when power is off!

Discharge with 20k-ohm resistor across terminals. Wait 30s, short again. Never use a screwdriver.

Testing

Visual: Bulging/leaking = replace | MFD: $\pm 10\%$ of rating | Ground: Terminal to case = OL (infinite)

HVAC School — Contactor & Capacitor Testing

6 Heat Pump & Defrost

Sequence & Troubleshooting

Heating Mode Sequence

1. Stat calls heat (W/Y energizes)
2. Reversing valve positions for heating (de-energized on most brands)
3. Compressor starts — outdoor coil = evaporator (absorbs heat)
4. Indoor coil = condenser (rejects heat)
5. Supplemental heat (W2) engages if needed

Defrost Cycle

Triggers when ice builds on outdoor coil (typically $< 40^{\circ}\text{F}$ outdoor). Defrost board reverses to cooling mode, outdoor fan stops, heat strips energize. Duration: 5-15 min.

Problem	Check
Never defrosts	Defrost sensor/board, reversing valve
Defrosts too often	Sensor location/accuracy
Won't exit defrost	Sensor stuck, board relay
Ice on coil	Low charge, low airflow, defrost fail

Rev. valve: Most brands energize O in cooling. Ruud/Rheem energize B in heating.

HVAC School — Heat Pump Defrost

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7 Electrical Formulas

Ohm's Law, Power & Voltage Drop

Ohm's Law

$$V = I \times R \quad I = V \div R \quad R = V \div I$$

Power (Single Phase)

$$P = V \times I \quad P = I^2 \times R \quad P = V^2 \div R$$

Three-Phase Power

$$P = V \times I \times 1.732 \times PF \quad (PF \text{ typically } 0.8-0.95)$$

Voltage Drop (Copper)

$$VD = 2 \times 12.9 \times I \times D \div CM$$

K=12.9 (copper) | I=amps | D=one-way feet | CM=circular mils

AWG	Ampacity (60°C)	CM
14	15A	4,107
12	20A	6,530
10	30A	10,380
8	40A	16,510
6	55A	26,240

Motor FLA estimate: 240V 1-phase $\approx$ HP $\times$ 4.5 | 3-phase $\approx$ HP $\times$ 2.5

Always verify against NEC Article 310 and local code.

NFPA 70 — National Electrical Code

8 Troubleshooting Flow

Decision Trees & PM Checklist

No Cooling

Stat on COOL, below room temp?

No → Check stat, batteries, display

Yes → Outdoor unit running?

No → Breaker → disconnect → contactor → cap → compressor

Yes → Check suction/head pressure

Low suction, low head → Low charge / leak

High suction, high head → Restricted airflow / overcharge

Low suction, high head → Restriction (filter drier, TXV)

High suction, low head → Bad compressor valves

Short Cycling

Check: stat location (drafts), filter, coil freeze-up, low charge, oversized equip, pressure switch

Pressure Washing PM

Cover electricals | $\leq$ 1500 PSI for condenser coils | Top-down spray | Acid cleaner for condenser, alkaline for evaporator | Rinse thoroughly | Dry before re-energizing | Straighten fins with comb

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OSHA Safety Standards